3.1.2 Amount of substance: Sheet 5

Titrations · 20 marks · about 25 minutes

Key information

- Mean titre: concordant titres only (within 0.10 cm^3); never include the rough.
- **n = c × V** with V in dm^3 ($cm^3 \div 1000$). Use the equation ratio, then scale up (eg 25.0 cm^3 from 250 cm^3 → × 10).
- Back titration: moles of excess reagent added – moles left over (from titration) = moles that reacted.
- Rinse burette and pipette with the solution they will hold; rinse the conical flask with distilled water.

Section A: Self-quiz

01 A student uses a volumetric flask to dilute 10 cm^3 of battery acid to 250 cm^3 with distilled water. 25.0 cm^3 samples are titrated with sodium hydroxide, but concordant results are not obtained.

Which suggestion would improve the chances of obtaining concordant titres?

[1 mark]

A Invert the volumetric flask several times after adding the distilled water.
B Wash the pipette with distilled water between each titration.
C Add extra drops of indicator to the sample when nearing the end point in each titration.
D Use a more concentrated solution of sodium hydroxide in the burette.

02 Which of these practical steps will improve the accuracy of a titration?

[1 mark]

A Using a 10.0 cm^3 pipette instead of a 25.0 cm^3 pipette.
B Rinsing the sides of the conical flask with water.
C Rinsing the burette with water before filling.
D Using 6 drops of indicator instead of 3 drops of indicator.

03 A student has a 10 cm^3 sample of 1.00×10^{-2} mol dm^{-3} methanoic acid solution. The student is asked to dilute it to a concentration of 2.00×10^{-4} mol dm^{-3} by adding distilled water. Which volume of water should be added?

[1 mark]

A 200 cm^3
B 490 cm^3
C 500 cm^3
D 510 cm^3

04 Calculate the mass, in g, of solid barium hydroxide (M_r = 171.3) needed to prepare 250 cm^3 of 0.100 mol dm^{-3} barium hydroxide solution.

[1 mark]

05 Give the meaning of the term concordant titres.

[1 mark]

Section B: Exam questions

06 A student is provided with a 5.60 g sample of ethanoic acid (CH_3COOH) contaminated with sodium ethanoate (CH_3COONa).

The student dissolves the sample in deionised water and makes the volume up to 200 cm^3

The student removes 25.0 cm^3 samples of the solution and titrates them with 0.350 mol dm^{-3} sodium hydroxide solution.

Table 1

	Rough	1	2	3
Final volume / cm^3	20.85	41.10	20.50	40.80
Initial volume / cm^3	0.00	20.85	0.00	20.50
Titre / cm^3	20.85	20.25	20.50	20.30

Use the results in Table 1 to calculate the mean titre value.

Use the mean titre to calculate the percentage by mass of sodium ethanoate in the original sample.

[6 marks]

07 Some tablets contain magnesium oxide. A student adds 25.0 cm^3 of 2.00 mol dm^{-3} hydrochloric acid (an excess, 0.0500 mol) to 6 tablets with a total mass of 2.14 g, and stirs until all the magnesium oxide has reacted.

$MgO + 2\ HCl \rightarrow MgCl_2 + H_2O$

The mixture is made up to 250 cm^3 with distilled water in a volumetric flask. 25.0 cm^3 of this diluted mixture is titrated with 0.0900 mol dm^{-3} sodium hydroxide solution.

$HCl + NaOH \rightarrow NaCl + H_2O$

Mean titre = 20.38 cm^3

Calculate the percentage by mass of magnesium oxide in the tablets.

[6 marks]

08 In an experiment, 0.952 g of solid barium hydroxide is used to make 250 cm^3 of standard barium hydroxide solution.

25.0 cm^3 of this barium hydroxide solution reacts with exactly 24.50 cm^3 of hydrochloric acid.

$Ba(OH)_2 + 2\ HCl \rightarrow BaCl_2 + 2\ H_2O$

Calculate the concentration of the hydrochloric acid. (M_r $Ba(OH)_2$ = 171.3)

[3 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**A**	Source: 7404/1 Jun 2018 Q20 (stem shortened)	1
02	**B**	Source: 7404/1 Jun 2024 Q23	1
03	**B**	Source: 7404/2 Jun 2018 Q09	1
04	0.100 × 250 ÷ 1000 × 171.3 = 4.28 g	Source: 7405/1 Jun 2023 Q06.1 Allow 4.3 g	1
05	Titres that are within 0.1 cm^3 of each other	Source: 7404/2 Jun 2017 Q05.3 Units are needed Allow 0.05–0.15 cm^3 Do not allow idea of identical results	1
06	M1 Mean titre = (20.25 + 20.30) ÷ 2 = 20.275 cm^3 M2 Moles NaOH = 0.350 × 20.275 ÷ 1000 = 0.00709625 mol (= moles CH_3COOH in 25.0 cm^3) M3 Moles CH_3COOH in 200 cm^3 = M2 × 8 = 0.05677 mol M4 Mass CH_3COOH = 0.05677 × 60.0 = 3.4062 g M5 Mass CH_3COONa = 5.6 – 3.4062 = 2.1938 g M6 % CH_3COONa = (2.1938 ÷ 5.6) × 100 = 39.1%	Source: 7404/1 Nov 2021 Q08.1 M6 allow 39.1 – 39.2% Allow ECF at each stage; accept alternative methods Alternative: M5 = % CH_3COOH = (M4 ÷ 5.6) × 100, then M6 = 100 – M5	6
07	M1 n(NaOH) = 0.0900 × 0.02038 = 0.0018342 mol (= n(HCl) in excess in 25 cm^3) M2 n(HCl in excess in 250 cm^3) = 0.0018342 × 10 = 0.018342 mol M3 n(HCl that reacted) = 0.0500 – 0.018342 = 0.031658 mol M4 n(MgO in 6 tablets) = 0.031658 ÷ 2 = 0.015829 mol M5 Mass of MgO = 0.015829 × 40.3 = 0.6379 g M6 % by mass of MgO = (0.6379 ÷ 2.14) × 100 = 29.8%	Source: 7404/1 Jun 2024 Q07.3 (stem shortened) M2 = M1 × 10; M3 = 0.0500 – M2; M4 = M3 ÷ 2; M5 = M4 × 40.3; M6 = M5 ÷ 2.14 × 100	6
08	M1 n $Ba(OH)_2$ in 25 cm^3 = (0.952 ÷ 171.3) × (25 ÷ 250) = 5.56 × 10^{-4} mol M2 n HCl in 24.5 cm^3 = 2 × 5.56 × 10^{-4} = 0.00111 mol M3 Concentration of HCl = 0.00111 × 1000 ÷ 24.50 = 0.045 mol dm^{-3}	Source: 7405/1 Jun 2023 Q06.5 (equation and M_r added from Q06.1) M2 = M1 × 2	3